

# Community-acquired methicillin-resistant *Staphylococcus aureus*: what do we need to know?

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Community-associated methicillin-resistant *Staphylococcus aureus* (CA-MRSA) has become a matter of concern worldwide, in particular in the USA. For the analysis of emergence and spread, clear definitions based on epidemiological origin are needed for discrimination between CA-MRSA, healthcare-associated community MRSA, and healthcare-associated MRSA (HA-MRSA). Although its role in pathogenesis is currently under debate, the capability for Panton-Valentine leukocidin formation is associated with the majority of CA-MRSA isolates from North America and from Europe. Most CA-MRSA isolates are attributed to clonal lineages different from HA-MRSA; there are, however, clonal lineages from which both HA-MRSA and CA-MRSA have been reported (e.g. ST1, ST5, ST8, and ST22); CA-MRSA ST8 (USA300), which is most frequent in the USA, has meanwhile been reported from Europe. CA-MRSA ST80 is widely disseminated in Europe; because of its pronounced oxacillin heteroresistance phenotype, ceftoxitin-based assays are advisable for reliable detection. So far, CA-MRSA infections seem to be much less frequent in Europe than in the USA, where patients with particular predispositions and low social status are at especial risk.